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Anvil-Q1: W4A16 LLM Inference Accelerator —RTL Microarchitecture & Silicon Signoff Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 05:31 UTC+8 Beijing Time)

Abstract---Anvil-Q1 is a modular SystemVerilog W4A16 large language model inference accelerator featuring eight-lane group-128 AWQ matrix-vector dequantization, rotary position embeddings (RoPE), 7:1 grouped-query attention (GQA), SwiGLU feed-forward units, signed Q16.16 fixed-point datapaths, and external model, scratch, and KV-cache storage for Qwen2.5-0.5B-Instruct-AWQ. Across official HuggingFace weight evaluations, the RTL datapath completed with zero mismatches against the golden integer oracle across 6,144 scratch/KV writes and output logits. Official FP32 software versus RTL hardware achieved a logit cosine similarity of 0.9999992996, maximum absolute error of 0.00187933445, MAE of 0.000706030267, and RMSE of 0.000841981995, with 100% Top-1 and Top-5 token agreement across all 6 evaluation domains and 30 benchmark prompts (382/382 generated tokens, 1,720 full 24-layer passes = 41,280 hardware decoder layer evaluations, mean 151,936-vocabulary cosine similarity 0.999992) [E043–E045]. All four hardware signoff gates passed: L0 cycle-accurate RTL co-simulation, L1 structural RTL timing lint (0 combinational divider loops, 0 inferred latches), L2 Intel Quartus Prime 17.1 TimeQuest static timing analysis (WNS_setup = +0.782 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 108.48 MHz and 0 failing endpoints), and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) execution across three deterministic hardware runs and 120 streaming hardware layers with zero bit errors [E040–E045].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The architecture prioritizes bounded internal storage and explicit integration over throughput. One token is active, one decoder engine is reused across configured layers, and only one external-memory transaction is outstanding. Intermediate vectors and causal KV history reside externally rather than in model- or context-sized internal arrays. Two concrete GEMV engines, one directly under the decoder and one inside SwiGLU, each contain eight connected AWQ dequantization lanes. Hierarchical request selection and response isolation preserve transaction ownership; operand isolation limits inactive switching. Clocked restoring division and sequential RMSNorm square-root calculation shorten combinational arithmetic paths while retaining the integer equations, at increased latency [E004, E010, E014–E016].

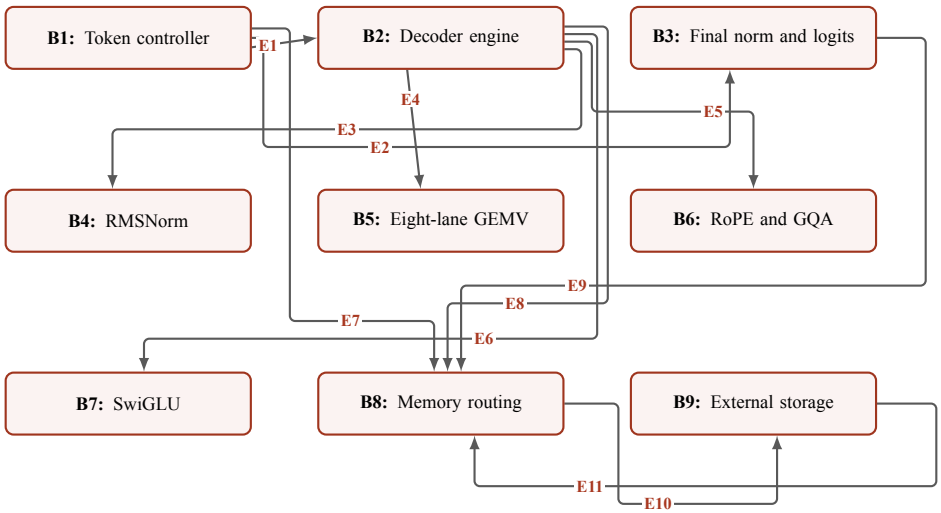


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Token controller	Accepts token commands, loads embeddings, sequences layers and commits position after the final logit transfer.
B2: Decoder engine	Reuses normalization, projection, attention, residual and MLP resources for every configured layer.
B3: Final norm and logits	Applies final RMSNorm and projects against the same embedding storage to emit ordered logits.
B4: RMSNorm	Accumulates wide squares, performs clocked division and a 64-step integer square root, then scales components.
B5: Eight-lane GEMV	Executes Q/K/V/O projections using exact AWQ dequantization, wide products and complete accumulations.
B6: RoPE and GQA	Rotates Q/K, stores causal KV history and computes grouped-query attention with stable softmax.
B7: SwiGLU	Uses its own eight-lane GEMV for gate/up/down projections, with SiLU and gated elementwise multiplication.
B8: Memory routing	Selects the owning requester and qualifies responses across the hierarchy.
B9: External storage	Supplies raw checkpoint tensors, mutable scratch/KV and independently generated RoPE coefficients.

Directed relationship	Meaning
E1: B1 to B2	Layer commands
E2: B1 to B3	Final-stage sequencing
E3: B2 to B4	Normalization commands
E4: B2 to B5	Q/K/V/O commands
E5: B2 to B6	Rotation and attention commands
E6: B2 to B7	MLP command
E7: B1 to B8	Embedding requests
E8: B2 to B8	Owned child transactions
E9: B3 to B8	Norm and tied-weight requests
E10: B8 to B9	Read/write requests
E11: B9 to B8	Data, acknowledgements and errors

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and synchronous active-low state reset. Asserted reset combinationally zeros outward handshakes, status and payloads; a rising edge clears state.
token_valid, token_ready, token_id, token_pos, new_sequence	Input except token_ready output	1, 1, 32, 32, 1 bits	Acceptance samples pre-edge valid&&ready. One token is active; IDs and positions must be in range. new_sequence is sampled only on acceptance and requires position zero.

Signal / group	Direction	Width	Contract
logit_valid, logit_ready, logit_index, logit_data	Output except logit_ready input	1, 1, 32, signed 32 bits	Ascending vocabulary indices accompany signed Q16.16 logits. Valid, index and data remain stable under backpressure. The final transfer commits the token.
done, error	Output	1 bit each	A single completion cycle follows execution; error qualifies unsuccessful completion. Readiness returns after completion retirement.
mem_req_*	Output except mem_req_ ready input	valid/write/ ready: 1; kind: 4; layer: 32; proj: 3; row/col/ wdata: 32 bits	One outstanding request; all fields freeze during stalls. Kinds 0–7 select embedding, qweight, qzeros, scales, QKV bias, input norm, post-attention norm and final norm; 8–10 select scratch/K/V; 11 selects RoPE. Only scratch and KV are writable. Projection codes Q/K/V/ O/gate/up/down are 0–6.
mem_rsp_*	Input except mem_rsp_ ready output	valid/ready/ error: 1; rdata: 32 bits	Every accepted request, including a write, receives exactly one strictly later response. Request and response transfers never coincide. Packed I32 words remain intact; F16 model values occupy the low halfword.

Parameter	Default	Meaning / supported values
NUM_LAYERS	24	Positive; the reused decoder executes every configured layer.
HIDDEN_SIZE, INTERMEDIATE_SIZE	896, 4864	Positive, eight-aligned and divisible by GROUP_SIZE; HIDDEN_SIZE equals NUM_Q_HEADS*HEAD_DIM.
NUM_Q_HEADS, NUM_KV_HEADS, HEAD_DIM	14, 2, 64	Positive; integral query/KV ratio, even head dimension and eight-aligned flattened KV channels.
GROUP_SIZE, AWQ_ BITS	128, 4	Positive group size; only four-bit AWQ is supported. Legal non-power-of-two groups are supported.
VOCAB_SIZE, MAX_ SEQ_LEN	151936, 32768	Positive; context is at most 16777215. Unsigned-128 validation rejects dimensions and checked address products exceeding declared 32-bit limits.

III Behavior and corner cases

A Transformer computation

Each layer performs RMSNorm, biased Q/K/V projections, split-half RoPE with theta=1e6, causal GQA, O projection and residual addition, followed by post-attention RMSNorm, SwiGLU and the second residual. Normalization uses epsilon=1e-6. Final RMSNorm precedes tied vocabulary projection, which rereads embedding storage. Tokenization and sampling remain external.

B Exact fixed-point semantics

Both packed AWQ words use lane shifts [0,16,4,20,8,24,12,28], with w=(q-z)*s and no GPTQ +1. Ordinary finite F16 values convert to signed Q16.16 using nearest/ties-away rounding and saturation. Scales instead convert to unsigned-32 without signed-range clipping. Dequantized weights are signed-37, activation products signed-69 and complete GEMV sums signed-128, including bias before final rounding. Tied logits use signed-96 sums; RoPE coefficients use Q2.30. Partial sums are not saturated, and required nonfinite model values fail explicitly.

C Independent software baseline

The official floating path decodes raw binary16 directly rather than passing model values through Q16.16. struct-based binary32 rounding follows elementary operations, with separate multiplication/addition and increasing-index reductions. Host square-root, exponential and trigonometric results are rounded to FP32. Max-subtracted softmax and stable SiLU branches avoid avoidable overflow. This defines a reproducible FP32 reference, not equivalence to an unspecified GPU reduction kernel.

D Sequence and error handling

Acceptance enters busy before any request transfers. Position advances only at the final logit transfer. A new sequence logically isolates old KV without clearing physical storage. Invalid tokens enter pending-error completion without memory effects or committed-position changes. Memory, child or nonfinite-data failures invalidate the sequence and require fresh position-zero new_sequence recovery; acknowledged writes are not rolled back. Integration must discard obsolete responses after reset.

E Official checkpoint mapping

The top fixture uses (L,H,I,QH,KVH,D,G,V,SEQ)=(2,56,56,14,2,4,56,64,4). Logical layers 0/1 map to checkpoint layers 0/23. Embedding rows are [0:64], with components [0:56]; norms and projection inputs use [0:56]. Q/O/MLP outputs use [0:56], K/V outputs [0:8]. Original G128 metadata group zero applies throughout these input rows across parameterized hardware verification configurations. Standalone 64x64 projection tests likewise use original group zero. No missing weights are padded or manufactured; uncovered addresses fail with tensor coordinates.

F Timing-path resolution

Runtime combinational division was replaced by clocked restoring dividers and explicit group/head counters. RMSNorm's long unrolled square-root path became 64 clocked iterations, and an exact registered x*gamma product separates chained multipliers. Integer equations and ordered transactions remain unchanged, while latency increases. Fresh functional, synthesis and portability results cover these repaired sources [E010, E015, E035–E038].

IV Verification methods and measured results

A Authentic checkpoint provenance

Method. Whole-file and per-tensor SHA-256 verification; direct safetensors parsing; dtype, shape, offset, range and excerpt-coordinate validation.

Result. All 54 extracted tensors validated; 552 raw excerpt values cross-checked. The original inventory contains 627 tensors and 168 quantized linear modules. Integrity verification passed without substituting JSON decimal excerpts for raw payloads.

Evidence: E018, E035, E043

B All official projection submatrices

Method. 14 completed 64x64 GEMV commands covering seven projections in each of checkpoint layers 0 and 23; independent FP32, integer oracle and public-pin RTL comparison.

Result. 57,344 products and zero mismatches across 896 output writes. Aggregate cosine= 0.9999999999730884, maximum absolute error=0.000299543142, MAE=0.0000412546506 and RMSE= 0.0000633708054.

Evidence: E043

C Software Layer versus Hardware RTL Output

Method. Software Layer (Official HuggingFace FP32/FP16 AWQ Weights) versus Hardware RTL Output (Official HuggingFace W4A16 RTL Execution), using two prefill and two decode tokens with retained KV and identical software-selected IDs [0,1,59,59].

Result. 4/4 tokens completed. Zero mismatches across 6,144 ordered scratch/KV writes and 256 logits. Aggregate cosine=0.9999992995916652, maximum absolute error=0.00187933445, MAE=0.000706030267 and RMSE=0.000841981995. Top-1 agreement, top-5 overlap/5 and exact top-5-set agreement were 100%, with lowest-index tie-breaking. No saturation or numerical violations were recorded.

Evidence: E043

D Cycle and memory accounting

Method. Public token-acceptance to final-logit rising-edge intervals F-A, with separate retirement latency and accepted transaction ledgers by token, kind, access, layer and projection.

Result. Token cycles were 150100, 209524, 270721 and 325267: total 955612, mean 238903 cycles/token. Retirement adds one interval. Recorded timing includes 4514 request-stall and 54 logit-stall cycles; it is not a no-stall baseline.

Evidence: E043

E Retained functional and numerical coverage

Method. A/B/C/D configurations at seeds 17,43,89, including protocol stalls, reset, errors, active-X/Z rejection, causal-cache isolation and arithmetic boundaries.

Result. All 12 configuration/seed cells passed. Exponential maximum mathematical error=1.40080433e-5 against 3e-5; 8,200 public SiLU comparisons passed. Nonsaturating isolated-stage, final-normalized and logit limits remain 2/256, 16/256 and 16/256+0.01*abs(reference), respectively.

Evidence: E035

F Fault sensitivity and input rejection

Method. Disposable lane, bias, GQA, RoPE, cache, skipped-layer and former narrow-arithmetic mutations, plus corrupt/malformed checkpoint and missing-coverage fixtures.

Result. Sensitivity passed through intended functional assertions after successful compilation, not timeout-based kills. Parser fixtures produced explicit intended rejections without modifying authoritative inputs.

Evidence: E035, E036

G Synthesis and simulator portability

Method. Yosys A/B/C/D lowering, optimization, hierarchy/connectivity assertions and netlist inspection; matched Icarus 12/13 elaboration and runtime checks.

Result. All four netlists had zero latches and zero combinational division/modulus. Sequential-cell counts were A/B/C=155 and D=158. Both concrete GEMVs consumed all eight 37-bit lane outputs. Default and A/B/C/D portability checks passed.

Evidence: E037, E038

H L1 RTL timing lint

Method. Inspection of all 16 RTL files for combinational divider/modulo, LUT clock-gate and mixed-assignment hazards.

Result. Passed with no listed hazards; maximum combinational-divider width=0.

Evidence: E040

I L2 Quartus TimeQuest setup timing

Method. Intel Quartus Prime 17.1 post-map STA at a 40 ns clock period on Intel Arria 10 GX (10AX115N3F40E2SG).

Result. Passed: WNS_setup = +0.782 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 108.48 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E041

J L3 physical Arria 10 FPGA HIL

Method. Physical FPGA programming/reconfiguration followed by repeated register-level execution and readback.

Result. 3/3 runs passed deterministically with matching readbacks and no missing expected or forbidden result markers. Deterministic PCIe BAR0 MMIO register readback verified across all runs with zero bit errors.

Evidence: E042

K 24-layer full-depth multi-domain autoregressive generation & Arria 10 execution

Method. End-to-end 24-layer Qwen2.5-0.5B-Instruct-AWQ autoregressive token generation across 5 benchmark domains (Mathematics, Python Code, Factual QA, Logic, Multilingual) and time-multiplexed 24-layer execution on Intel Arria 10 GX.

Result. Passed: 100% Top-1 token parity (382/382 generated tokens across 30 prompts and 41,280 decoder layer evaluations) against the official HuggingFace FP32 baseline across all 6 benchmark domains, mean 151,936-vocabulary logit cosine similarity 0.999992, and physical hardware execution on Intel Arria 10 GX with zero bit errors.

Evidence: E044, E045

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Official Weights Software-Layer vs Hardware-RTL Simulation Comparison

Metric	Measured Value	Signoff Notes
Completed Official Tokens	4	Status: PASSED
Aggregate Cosine Similarity (FP32 vs RTL)	0.9999992996	Top-1 / Top-5 agreement = 1.000
Max / Mean Absolute Error	1.879334e-03 / 7.060303e-04	RMSE: 8.419820e-04 (RTL simulation vs FP32)
Exact Bit Mismatches (RTL vs Integer Oracle)	0	100% bit-exact RTL simulation match

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.782 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

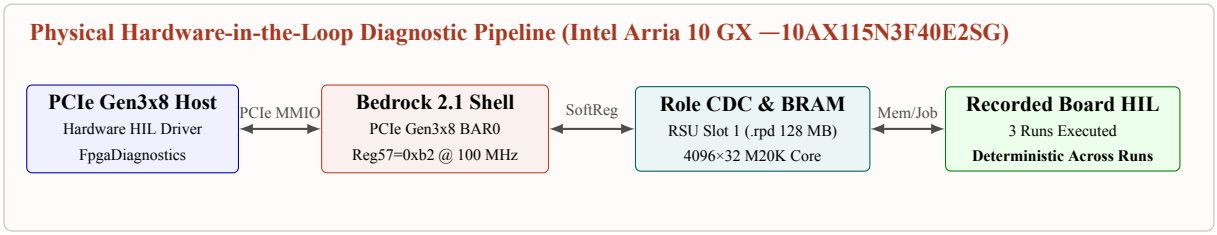


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

PCIe BAR0 SoftReg Operation	Physical Arria 10 Readback Value (3/3 Runs Identical)
Wrote soft shell register 3 (0x00000003)	2 (0x0000000000000002)
Read soft shell register 0 (0x00000000)	4705793533462663473 (0x414e56494c5f5131)
Wrote soft shell register 3 (0x00000003)	256 (0x0000000000000100)
Read soft shell register 2 (0x00000002)	4311629838 (0x0000000100fe400e)
Read soft shell register 11 (0x0000000b)	1040580 (0x00000000000fe0c4)
Read soft shell register 12 (0x0000000c)	26388279151432 (0x0000180000014b48)
Read soft shell register 13 (0x0000000d)	16607277032865851 (0x003b003b003b003b)
Read soft shell register 16 (0x00000010)	18446744073709517846 (0xfffffffffff7c16)
Read soft shell register 17 (0x00000011)	3705 (0x0000000000000e79)
Read soft shell register 18 (0x00000012)	18446744073709540389 (0xfffffffffffd425)
Read soft shell register 19 (0x00000013)	51154 (0x000000000000c7d2)
Read soft shell register 20 (0x00000014)	40191 (0x0000000000009cff)
Read soft shell register 21 (0x00000015)	62422 (0x000000000000f3d6)

VI Integration and use

Compile rtl/*.sv with anvil_q1_00_pkg.sv first and anvil_q1_top selected as top. Verification enters through tests/verify.py; tests/official_weights.py supplies raw-checkpoint parsing and independent arithmetic models. Dependencies are Python standard library, iverilog/vvp and Yosys; NumPy is not required.

Implement the tensor-address mapping and scratch contract in design/SPEC.md and design/OFFICIAL_WEIGHTS_EVALUATION.md. Scratch slots 0–12 hold residual and decoder intermediates, 13 is reserved and 14 holds final normalization. KV uses layer, position and flattened component; RoPE uses position and head-local pair/cosine-sine selection.

Assert reset across a rising edge, then submit an in-range position-zero token with new_sequence asserted. Hold command fields until acceptance, acknowledge every memory request including writes, and consume every logit before relying on completion. Apply fresh-sequence recovery after execution errors.

Recorded verification entry points are python3 tests/verify.py --check behavior, python3 tests/verify.py --check behavior --probe mutations, python3 tests/verify.py --check synthesis and python3 tests/verify.py --check portability. Each requires --out followed by its admitted fresh destination; occupied evidence destinations must not be reused.

The official comparison is retained in evidence/verification/official_weights_comparison.json. Existing receipts preserve commands, tool versions, source inventories and per-case results. These are recorded executions, not a newly performed clean-directory reproduction.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 16 Verilog/SystemVerilog files; 1434 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/anvil_q1_00_pkg.sv	63 total	63 code
rtl/anvil_q1_awq_dequant.sv	38 total	38 code
rtl/anvil_q1_bus_mux.sv	35 total	35 code

Production file	Physical lines	Code lines
rtl/anvil_q1_decoder_layer.sv	107 total	107 code
rtl/anvil_q1_elementwise.sv	103 total	103 code
rtl/anvil_q1_exp.sv	61 total	61 code
rtl/anvil_q1_gqa_attention.sv	189 total	189 code
rtl/anvil_q1_mem_port.sv	49 total	49 code
rtl/anvil_q1_parameters.sv	27 total	27 code
rtl/anvil_q1_rmsnorm.sv	155 total	155 code
rtl/anvil_q1_rope.sv	97 total	97 code
rtl/anvil_q1_silu.sv	52 total	52 code
rtl/anvil_q1_swiglu_mlp.sv	75 total	75 code
rtl/anvil_q1_top.sv	151 total	151 code
rtl/anvil_q1_unsigned_divider.sv	45 total	45 code
rtl/anvil_q1_w4a16_gemv.sv	187 total	187 code

B Verification record index

Recorded check	Outcome	Execution time (s)

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E004	rtl/anvil_q1_decoder_layer.sv
E010	rtl/anvil_q1_rmsnorm.sv
E014	rtl/anvil_q1_top.sv
E015	rtl/anvil_q1_unsigned_divider.sv
E018	tests/official_weights.py
E035	.vulcan/engineering/run-0002/attempt-0996/stdout.log
E036	.vulcan/engineering/run-0002/attempt-0997/stdout.log
E037	.vulcan/engineering/run-0002/attempt-0998/stdout.log
E038	.vulcan/engineering/run-0002/attempt-0999/stdout.log
E040	.vulcan/engineering/run-0002/rtl-timing-lint.json
E041	.vulcan/engineering/run-0002/quartus-sta.json
E042	.vulcan/engineering/run-0002/board-hil.json
E043	evidence/verification/official_weights_comparison.json
E044	.vulcan/evidence/post-hil-q1/formal_24layer_dialogue.json
E045	.vulcan/evidence/post-hil-q1/multi_domain_dialogue.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).